

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0392

Evergreen Park 230 kV
March 19, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0392. The Interconnection Customer proposes a 297.1 MW natural gas combined cycle facility interconnecting at the Evergreen Park 230 kV substation in the Illinois area. The request was received on October 15, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 216 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,331,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FISK ST – EVERGREENPARK 230.0 kV	EVERGREEN PARK – SPRING CITY 230.0 kV line	123.4	930	51.1
SPRING CITY – EVERGREEN PARK 230.0 kV	EVERGREEN PARK – FISK STREET 230.0 kV	111.0	990	55.1
CONESVILLE STA – DRESDEN 500.0 kV	DRESDEN – PT ARENA 500.0 kV	109.8	2,422	6.1
SPRING CITY – WASHINGTON PARK 230.0 kV	EVERGREEN PARK – FISK STREET 230.0 kV Line	102.2	876	32.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FISK ST – EVERGREENPARK 230.0 kV	Rebuild line with higher-capacity conductor	\$23,894,000	17% (\$3,985,000)
SPRING CITY – EVERGREEN PARK 230.0 kV	Rebuild line with higher-capacity conductor	\$9,800,000	17% (\$1,662,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CONESVILLE STA – DRESDEN 500.0 kV	Reconductor line	\$13,172,000	2% (\$309,000)
SPRING CITY – WASHINGTON PARK 230.0 kV	Reconductor line	\$2,648,000	14% (\$375,000)
	Total allocated to project		\$6,331,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.